

Harsha Prada Chandrakumar

Full Stack Developer | AI/ML & GenAI Engineer

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EDUCATION

Kumaraguru College of Technology Bachelor of Technology in Artificial Intelligence and Data Science - 9.32 **Sept 2023- May 2027**
Bharini Vidhyalaya Senior Secondary School Higher Secondary Education CBSE – 97.4% **May 2023**
Bharini Vidhyalaya Senior Secondary School Senior Secondary Education CBSE – 94.8% **May 2021**

SKILLS

Languages: *Proficient* Python(3yrs) · SQL(3yrs) · JavaScript(2yrs) *Intermediate* C++(<1yrs)
Full-Stack Development: HTML · CSS · JavaScript · Node.js · Express.js · Django
Databases: MongoDB · SQLite · MySQL · Firebase · Postgres
Developer Tools Git · GitHub · Docker · Netlify · Vercel · Termius · Sentry
Machine Learning & Deep Learning: Regression · Classification · Clustering · CNNs · LSTMs · YOLO
AI/ML Tools: TensorFlow · Scikit-learn · OpenCV · Numpy · Pandas
Generative AI: Prompt Engineering · Diffusion Models · Large Language Models (LLMs)
Problem Solving (Data Structures & Algorithms, CodeChef, LeetCode)

EXPERIENCE

iQube INNOVATION CENTRE, Kumaraguru College of Technology(iqubekct.ac.in) **Coimbatore, IN**
WEB DEVELOPER – LEVEL 2 TECHNICAL MEMBER **Oct 2024 – Present**

Kash (Blockchain-Backed FinTech Platform) | kash.kact.ac.in

- Developed a Django/PostgreSQL fintech system for **3,000+ users** with blockchain-backed transaction logging.
- Built REST APIs and Dockized services to reduce audit effort by **75%** across financial workflows.
- Designed scalable backend workflows and monitoring dashboards for real-time financial data systems

Yugam Fest Platform (Large-Scale Event Platform) | yugam.in

- Developed a Django-based event system for **10,000+ users** across **100+ events** with Razorpay payment integration.
- Optimized backend throughput by 40% using PostgreSQL tuning and Redis caching for high-concurrency workloads.
- Implemented dashboards, data pipelines, and WhatsApp API bot integrations for automated event operations.

SAMSUNG PRISM RESEARCH INTERNSHIP **Coimbatore, IN**
AI RESEARCH ENGINEER INTERN – MULTI-AGENT SYSTEMS & GENAI **Feb 2025 - Present**

Project-Based Study (PBS): Cross-Agent Contextual Information Storage and Retrieval

- Developed a multi-agent conversational system for context-aware collaboration and intelligent knowledge retrieval.
- Built autonomous agents using LangChain, LangGraph, CrewAI, and AstraDB with long-term memory and modular routing.
- Integrated a RAG pipeline with real-time Wiki search for knowledge grounding and persistent contextual reasoning.
- Explored LLM orchestration, multi-agent coordination, memory architectures, and advanced prompt engineering systems.

IEEE COMPUTER SOCIETY, KCT **Coimbatore, IN**
STUDENT MEMBER **Feb 2024 - Present**

- Published 2 IEEE–IITDM papers:
 - “Performance Analysis of Open-Source Diffusion Models for Artistic Image Generation”**
 - “Multilingual Sentiment Analysis Framework for Video News Verification Using AI”**
- Organized hackathons, ideathons, sustainathons, and AI workshops for campus innovation.
- Worked on GenAI research in misinformation detection and multilingual media analytics.

PROJECTS

ResumeXpert – AI-Powered Resume Builder (~40 hours) | Firebase, Node.js, Express.js, MongoDB, HTML, CSS, JavaScript

- Built a smart resume builder using Firebase, Node.js, and Gemini API for AI-enhanced resume generation and PDF export.
- Included dynamic sections, user authentication, and MongoDB integration for personalized dashboards.

Code Quality Review Agent (~24 hours) | Python, Node.js, ESLint, Pylint, Radon, Gemini API

- Built an editor- and CI-ready code review agent using Python, Node.js, and Gemini API to automate code quality analysis.
- Integrated ESLint, Pylint, and Radon to generate structured JSON reports with scoring and AI-driven improvement suggestions.

Liver Segmentation – AI for Medical (~24+ hours) | Python, PyTorch, U-Net, OpenCV, NumPy, Medical Decathlon Dataset, Matplotlib

- Applied U-Net based deep learning on CT scans from the Medical Decathlon dataset to detect liver cancer.
- Built for the CodHer Hackathon to enhance early diagnosis through AI-powered image segmentation

Open Day Project – Interactive College Web Portal (~12 hours) | HTML, CSS, JavaScript

- Built a college portal using HTML, CSS, JS, and OpenStreetMap for events, departments, and campus navigation.
- Enabled smooth exploration of college features and map-based virtual campus experience.

PosturePerfect – IoT-Based Posture Monitoring System (~24 hours)

- Developed a wearable solution using MPU6050, Flex Sensor, and RFID for real-time posture tracking and alerting.
- Presented at an Ideathon to promote ergonomic wellness through sensor-based monitoring.

HACKATHON

2nd Place(College) –CPP Coding Contest | (Global Coding Challenge)

Nov 2025

111 Countries | 3000 Universities Worldwide | 2nd in College | Nov 2024

- **Problem Solved:** Solved advanced algorithmic challenges in a collaborative team setting, demonstrating strong problem-solving and time-critical coding skills.

4th Place – IEEE Xtreme 17.0 | (24-Hour Global Coding Challenge)

Jan 2025

120+ Countries | Top 8000 Teams Worldwide | 4th in College | Oct 2024

- **Problem Solved:** Solved complex algorithmic problems in a team setting under strict time constraints. Demonstrated advanced coding and collaboration skills across 24 hours.

Finalist – Vision X Hackathon | (AI + Web Development)

Nov 2025

Source Code Analysis Project | Selected For Internship | Nov 2025

- **Problem Solved:** Built an automated AI-powered code review agent that analyzes source quality and generates actionable improvement suggestions in real time.

Finalist – ACM CodHer Hackathon | (AI/ML + Computer Vision)

Dec 2024

Liver Segmentation Project | 24-Hour Hackathon | Nov 2024

- **Problem Solved:** Built a deep learning model for liver segmentation using the Medical Decathlon dataset to automate medical image analysis.

Awarded – Best Innovation Team – Designathon | (IoT Track)

Nov 2024

Posture Perfect Wearable | 30-Hour Innovation Sprint | Aug 2024

- **Problem Solved** Designed and prototyped a wearable for posture correction using MPU6050, RFID, and flex sensors. Recognized for innovation and execution.

Project Showcase – Open Day Hackathon | (~12-Hour Build)

Feb 2025

Real-Time Web App with OpenStreetMap | Apr 2024

- **Problem Solved:** Developed a cab and cricket ticket booking system using HTML, CSS, JS, and OpenStreetMap API. Showcased to faculty and peers during the open day.

CERTIFICATIONS

- **Advanced MERN Stack Development** – Infosys Springboard
- **Project-Based Learning (PBL) Student Certification-** Collision Avoidance Using Graph Theory(1st year)
- **Python Development** – Udemy
- **AI & ML Internship** – InternPe
- **AICTE Virtual Internship on AI/ML with Alteryx**
- **AI & ML with Computer Vision** – CodHer Hackathon Certification
- **IoT Design Thinking** – Designathon Certification

VOLUNTEERING

Tech Support Team – Nilgiris Hackathon, Kumaraguru College of Technology | Feb 2026

Delivered hands-on technical mentoring at hackathons and ideations, supporting teams with architecture decisions, coding challenges, and innovation evaluation.

Volunteer – IEEE Student Branch, Kumaraguru College of Technology | Feb 2024 – Present

Organized hackathons, Ideathons, and coding events, managing logistics, mentorship coordination, and event promotions to boost student innovation in emerging tech.

Volunteer – Campus Placement Drive Support, KCT

Assisted recruiters and coordinated student participation, ensuring smooth scheduling, interview flow, and an efficient end-to-end placement process.